

# Nima Nouri

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## Research Interests

Reinforcement Learning • Intelligent Control • Robotics & Mechatronics  
Solar Energy Systems • Solar Tracking & Backtracking • Autonomous Systems  
Mechanism Design • Embedded Systems & IoT

## Education

**B.Sc. in Mechanical Engineering**

2023 – Present

Sharif University of Technology

**GPA: 18.24/20 | Ranked in the Top 15% of Mechanical Engineering Students**

## Research Engineering Projects

### Solar PV Tracking and Backtracking Mechanism Research

2026 – Present

Research on horizontal single-axis PV trackers, backtracking, inter-row shading, sloped terrain, and mechanically coordinated single-motor tracker architectures.

Supervisor: [Supervisor Name]

- Investigating cam, linkage, and transmission-based mechanisms for coordinated tracking and backtracking.
- Conducting literature review and concept development for low-complexity solar tracker systems.

### Modeling and Control of an Inverted Pendulum

2026

Developed a cart-pole model using MATLAB/Simulink and Simscape Multibody and designed its control system.

Supervisor: [Supervisor Name]

- Designed an LQR controller for upright stabilization.
- Implemented energy-based swing-up control and switching between swing-up and stabilization modes.

### Arduino-Based Step Counter

2026

Developed an embedded motion-sensing system for step detection and real-time sensor-data processing.

Supervisor: [Supervisor Name]

- Integrated motion sensors with Arduino and implemented threshold-based motion and step detection.

### Telescopic Jib Crane for a 3-Ton Forklift

2025

Designed an adjustable telescopic jib crane with an approximate rated lifting capacity of 1.5 tons.

Supervisor: [Supervisor Name]

- Performed structural calculations and developed the complete CAD assembly using SolidWorks.

### Modeling and Analysis of MDOF Dynamic Systems

2024

Developed mathematical and numerical models of multi-degree-of-freedom mechanical systems in MATLAB.

Supervisor: [Supervisor Name]

- Analyzed natural frequencies, mode shapes, free vibration, forced vibration, and transient response.

### Mechanical Component and Power Transmission Design

2024 – 2025

Designed and analyzed mechanical components and power-transmission systems.

Supervisor: [Supervisor Name]

- Designed shafts, springs, bearings, gears, gearboxes, bolted joints, and welded structures.

### Computational Fluid Dynamics Projects

2024

Performed numerical analysis of internal and external fluid-flow problems using ANSYS Fluent.

Supervisor: [Supervisor Name]

- Simulated airfoil aerodynamics and internal pipe-flow problems.

## Technical Skills

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|-----------------------|--------------------------------------------------------------------------------------------|
| Programming & Control | Python, MATLAB, Simulink, Classical Control, State-Space, PID, LQR, Reinforcement Learning |
| Modeling & Simulation | Simscape Multibody, MSC Adams, ANSYS Mechanical, ANSYS Fluent                              |
| Mechanical Design     | SolidWorks, AutoCAD, Machine Design, Mechanism Design, Structural Analysis                 |
| Embedded & IoT        | Arduino, Sensor Integration, Data Acquisition, IoT Fundamentals                            |
| Networking            | Network+ / CCNA Fundamentals, TCP/IP, Switching, Routing, Subnetting and VLANs             |
| Tools                 | Git, LaTeX, Microsoft Office                                                               |

## Selected Coursework

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Automatic Control • Mechanical Vibrations • Dynamics • Machine Design  
Fluid Mechanics • Heat Transfer • Numerical Methods • Computer Programming

## Athletics & Extracurricular Activities

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### Competitive Football Goalkeeper

- Competitive goalkeeper since the age of 13, with experience in Iran's Division 2, Division 1, and Premier League competitions.
- Played for clubs including Javid Karaj, Moqavemat, and Piyous.

## Languages

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|----------|----------------------------------|
| Persian  | Native                           |
| English  | Professional Working Proficiency |
| Japanese | Beginner                         |

## References

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*References available upon request.*